



Probabilistic Forecasting of Electricity Prices

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<https://nikum.org/>

Challenges in Forecasting Electricity Prices

Day-ahead electricity prices from Germany, 2015/01/01 - 2020/12/31.

Auction takes place at noon, all hours of the following day are traded in an uniform price auction.

- High volatility and seasonality.
- Influence of external factors.
- Complex dependency structures.
- Choice of probabilistic model.
- Variable selection.
- Evaluation and interpretation of forecasts.

Dataset (Preliminary)

P: Prices (EUR/MWh)

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-130.09	25.92	34.02	34.57	43.59	200.04

L: Load DA forecast

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
28824	47044	54776	54852	62982	75912

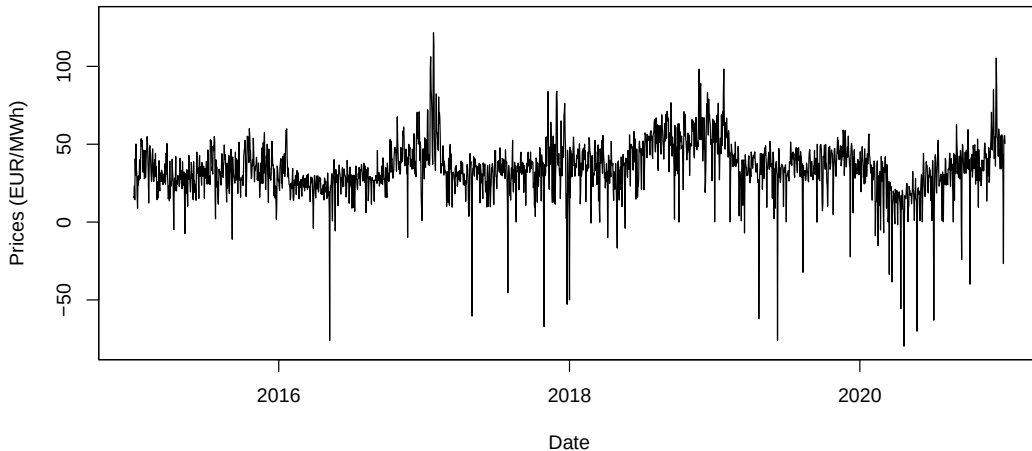
R: Renewables DA forecast:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
574.2	7907.8	14437.8	16309.9	23086.1	62490.1

E: EUA; C: API2 coal; G: TTF gas; O: Brent oil.

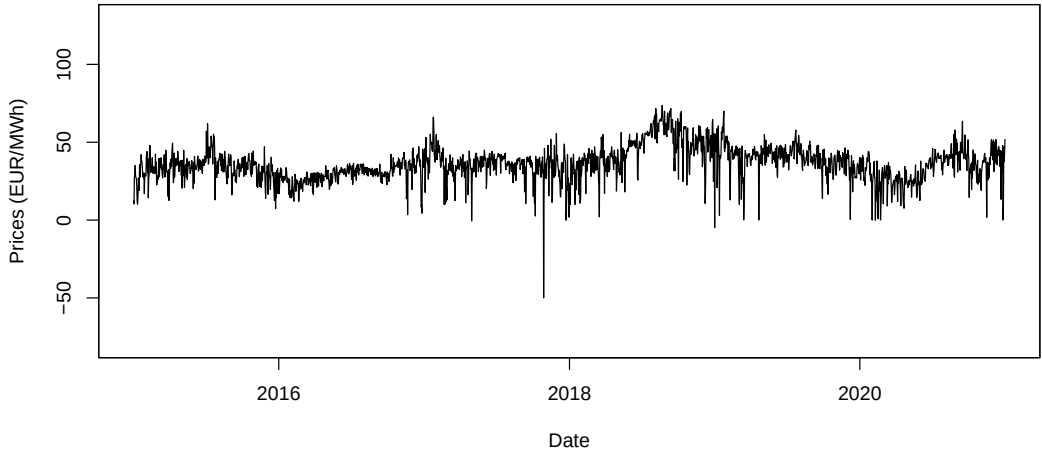
Hourly Prices

Hour 12, raw data:



Hourly Prices

Hour 22, raw data:



Regression

A typical regression model for electricity price forecasting

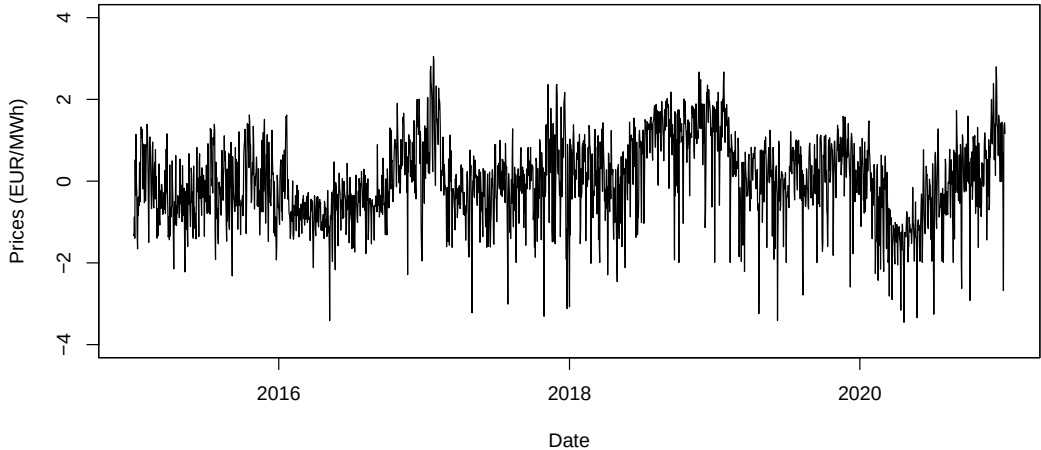
$$P_{d,h} = \beta_0 + \beta_1 P_{d-1,h} + \beta_2 L_{d,h} + \beta_3 R_{d,h} + \dots + \varepsilon_{d,h}, \quad \varepsilon_{d,h} \sim N(0, \sigma^2)$$

where:

- d, h : The day and hour.
- $P_{d,h}$: Electricity price (usually transformed).
- $P_{d-1,h}$: Electricity price of previous day (lagged price).
- $L_{d,h}, R_{d,h}$: Load and renewable forecast.
- $\dots$: Seasonal factors (e.g., time of day, day of week), other lagged variables E, C, G, O.
- $\beta_0, \beta_1, \beta_2, \beta_3, \dots$: Coefficients to be estimated.

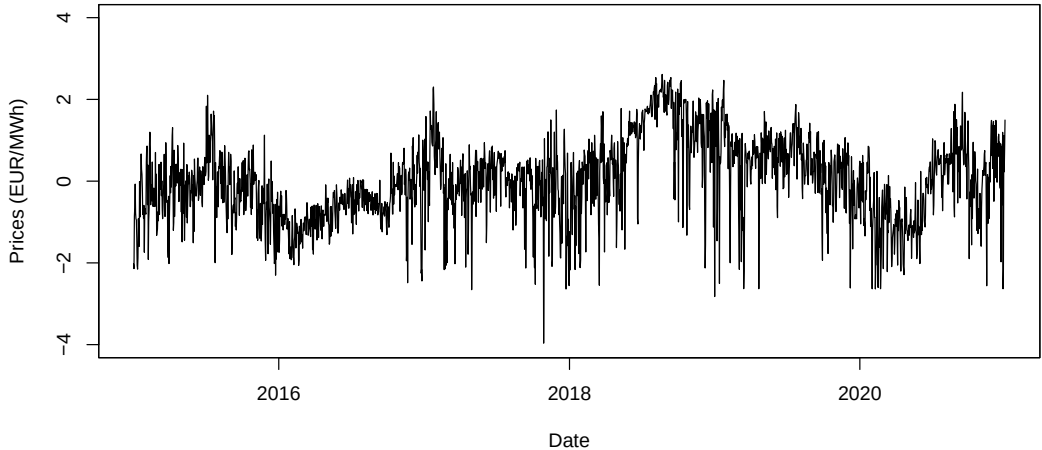
Hourly Prices

Hour 12, transformed data:



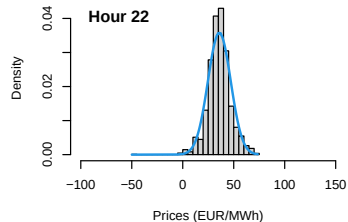
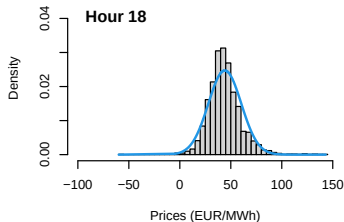
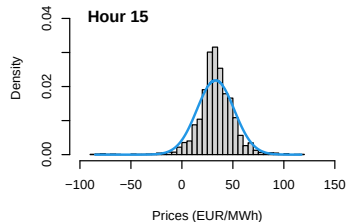
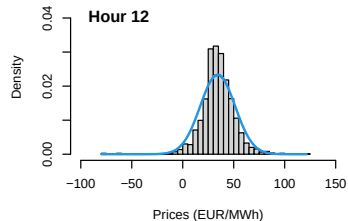
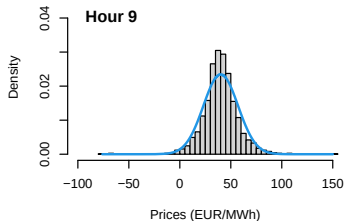
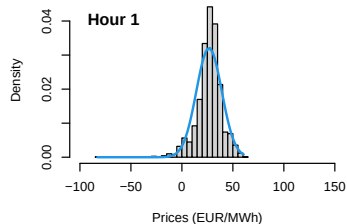
Hourly Prices

Hour 22, transformed data:



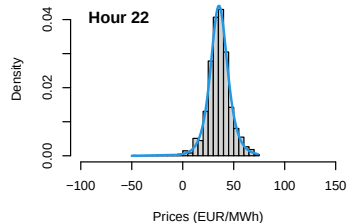
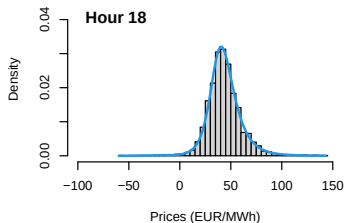
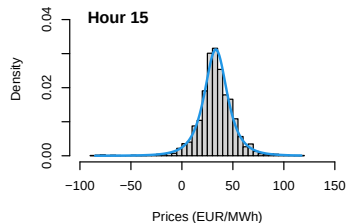
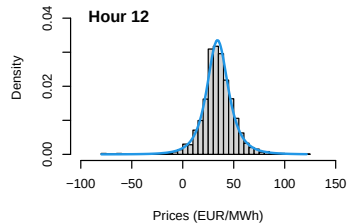
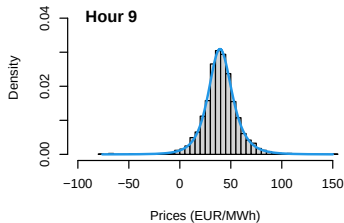
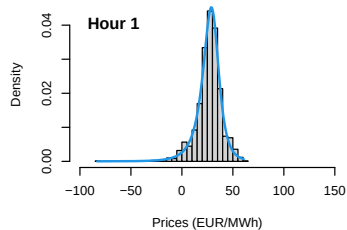
Hourly Distributional Models

Normal distribution:



Hourly Distributional Models

Johnson's SU- distribution:

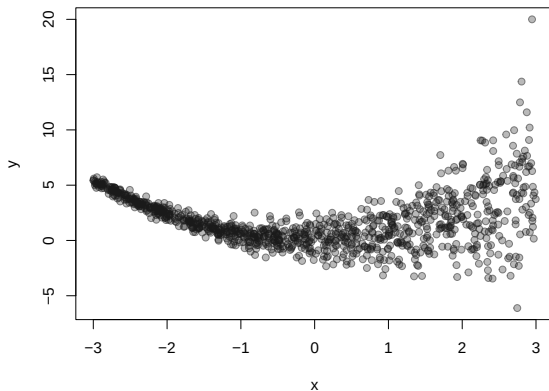


Regression Beyond the Mean

- Classical regression has focused on relating the conditional mean of a response y_i to covariate information x_i for observations $(x_1, y_1), \dots, (x_n, y_n)$.
- Linear model:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

with ε_i i.i.d. $N(0, \sigma^2)$



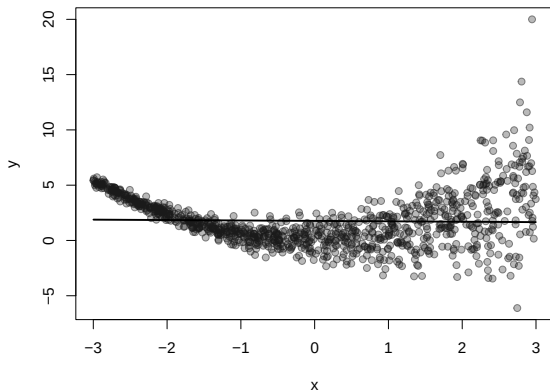
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$$\Rightarrow E(y_i | x_i) = \mu_i(x_i) = \beta_0 + \beta_1 x_i$$



Regression Beyond the Mean

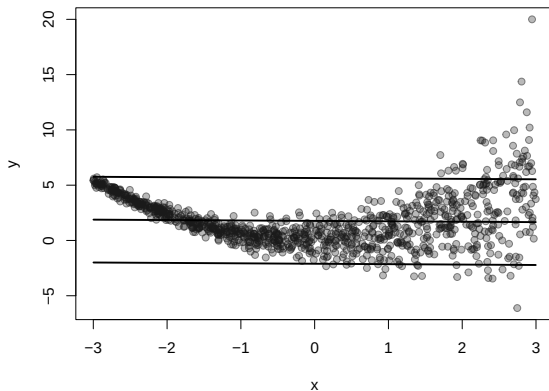
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with ε_i i.i.d. $N(0, \sigma^2)$

$$\Rightarrow E(y_i | x_i) = \mu_i(x_i) = \beta_0 + \beta_1 x_i$$

$$\Rightarrow \text{Var}(y_i | x_i) = \sigma^2$$

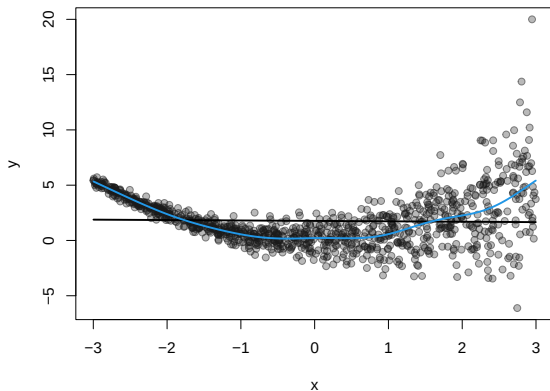


Regression Beyond the Mean

- Classical regression has focused on relating the conditional mean of a response y_i to covariate information x_i for observations $(x_1, y_1), \dots, (x_n, y_n)$.
- Nonparametric model

$$E(y_i|x_i) = \mu_i(x_i) = \beta_0 + f(x_i)$$

$$\text{with } y_i|x_i \sim N(\mu_i(x_i), \sigma^2)$$



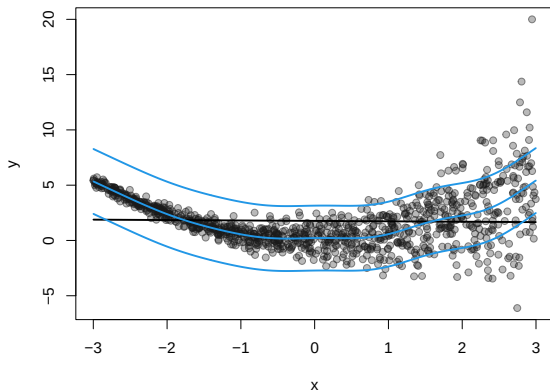
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σ^2 fixed.



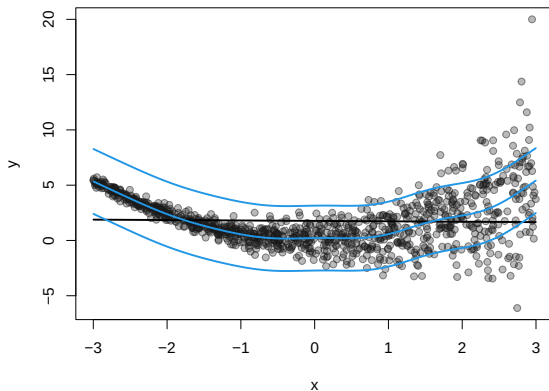
Regression Beyond the Mean

- Nonparametric model for location and scale.

- $E(y_i|x_i) = \mu_i(x_i) = \beta_0^\mu + f^\mu(x_i)$

$$\begin{aligned}\text{Var}(y_i|x_i) &= \sigma_i^2(x_i) \\ &= \exp\left(\beta_0^{\sigma^2} + f^{\sigma^2}(x_i)\right)\end{aligned}$$

with $y_i|x_i \sim N(\mu_i(x_i), \sigma_i^2(x_i))$.



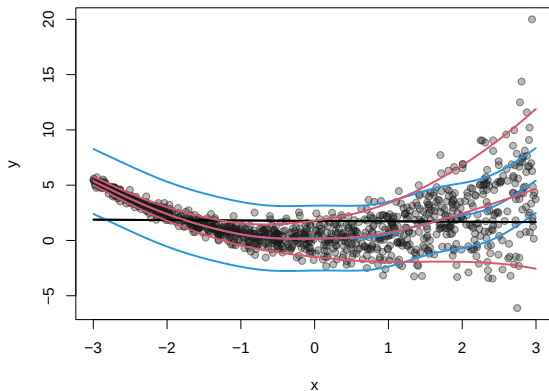
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with $y_i|x_i \sim N(\mu_i(x_i), \sigma_i^2(x_i))$.



Model Specification

Any parameter of a population distribution $\mathcal{D}$ may be modeled by explanatory variables

$$y \sim \mathcal{D}(\theta_1(\mathbf{x}; \beta_1), \dots, \theta_K(\mathbf{x}; \beta_K)),$$



with $\beta = (\beta_1^\top, \dots, \beta_K^\top)^\top$.

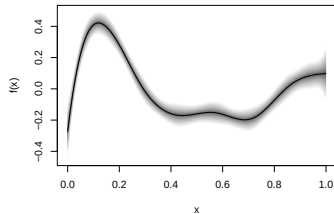
Each parameter is linked to a structured additive predictor

$$h_k(\theta_k(\mathbf{x}; \beta_k)) = f_{1k}(\mathbf{x}; \beta_{1k}) + \dots + f_{j_k k}(\mathbf{x}; \beta_{j_k k}),$$

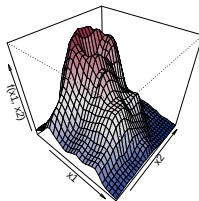
- $j = 1, \dots, J_k$ and $k = 1, \dots, K$.
- $h_k(\cdot)$: Link functions for each distribution parameter.
- $f_{jk}(\cdot)$: Model terms of one or more variables.

Model Terms $f_{jk}(\cdot)$

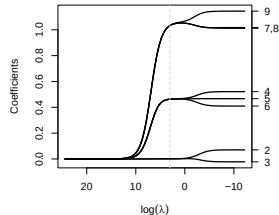
Nonlinear Effects



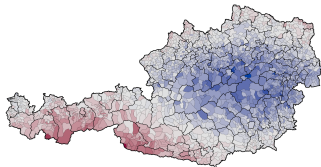
Two-Dimensional Surfaces



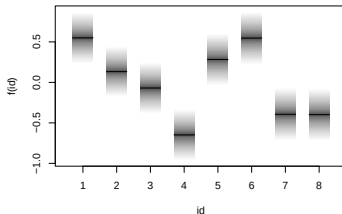
LASSO & Factor Clustering



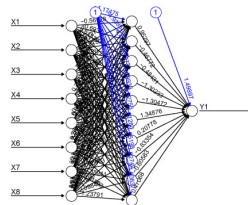
Spatially Correlated Effects $f(x) = f(s)$



Random Intercepts $f(x) = f(id)$



Neural Networks



Estimation

The main building block of regression model algorithms is the probability density function $d_y(\mathbf{y}|\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_K)$.

Estimation typically requires to evaluate the log-likelihood

$$\ell(\boldsymbol{\beta}; \mathbf{y}, \mathbf{X}) = \sum_{i=1}^n \log d_y(y_i; \theta_1(\mathbf{x}_i; \boldsymbol{\beta}_1), \dots, \theta_K(\mathbf{x}_i; \boldsymbol{\beta}_K)),$$

with $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_K)$.

The log-posterior (frequentist penalized log-likelihood)

$$\log \pi(\boldsymbol{\beta}, \boldsymbol{\tau}; \mathbf{y}, \mathbf{X}, \boldsymbol{\alpha}) \propto \ell(\boldsymbol{\beta}; \mathbf{y}, \mathbf{X}) + \sum_{k=1}^K \sum_{j=1}^{J_k} [\log p_{jk}(\boldsymbol{\beta}_{jk}; \boldsymbol{\tau}_{jk}, \boldsymbol{\alpha}_{jk})],$$

where $p_{jk}(\cdot)$ are priors, $\boldsymbol{\tau}_{jk}$ (smoothing) variances and $\boldsymbol{\alpha}_{jk}$ fixed hyper parameters.

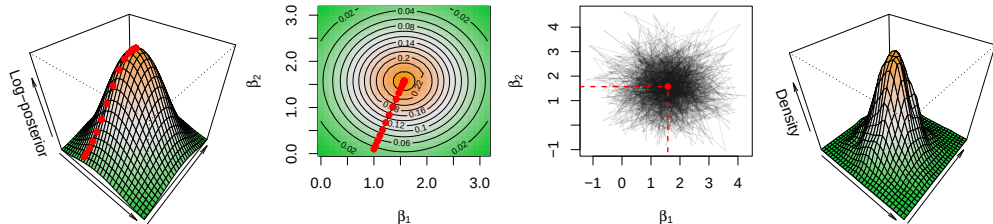
Estimation

Bayesian point estimates of parameters are obtained by:

- 1 Maximization of the log-posterior for posterior mode estimation.
- 2 Solving high dimensional integrals, e.g., for posterior mean or median estimation.

Problems 1 and 2 are commonly solved by computer intensive iterative algorithms of the following type:

$$(\beta^{[t+1]}, \tau^{[t+1]}) = U(\beta^{[t]}, \tau^{[t]}; \mathbf{y}, \mathbf{X}, \alpha).$$



L1-Type Penalization

Idea: depending on the type of covariate effects, subtract a combination of (parts of) the following penalty terms $\tau^{-1}J(\beta)$ from the log-likelihood.

Classical LASSO (Tibshirani, 1996): For a metric covariate x_{jk} use

$$J_m(\beta_{jk}) = |\beta_{jk}|.$$

Group LASSO (Meier et al., 2008): For a (dummy-encoded) categorical covariate $\mathbf{x}_{jk}$ use

$$J_g(\beta_{jk}) = \|\beta_{jk}\|_2,$$

with vector β_{jk} collecting all corresponding coefficients.

L1-Type Penalization

Alternatively, for categorical covariates often *clustering* of categories with implicit *factor selection* is desirable.

Fused LASSO (Gertheiss and Tutz, 2010): Depending on the *nominal* (left) or *ordinal* scale level (right) of the covariate, use

$$J_f(\beta_{jk}) = \sum_{l>m} w_{lm}^{(jk)} |\beta_{jkl} - \beta_{jkm}| \quad \text{or} \quad J_f(\beta_{jk}) = \sum_{l=1}^{c_{jk}} w_l^{(jk)} |\beta_{jkl} - \beta_{jk,l-1}|$$

where c_{jk} is the number of levels of categorical predictor $\mathbf{x}_{jk}$ and $w_{lm}^{(jk)}$, $w_l^{(jk)}$ denote suitable weights. Choosing $l = 0$ as the reference, $\beta_{jk0} = 0$ is fixed.

New: Fused LASSO for metric covariates incl. splines for nonlinear effects.

L1-Type Penalization

Quadratic approximations of the penalties (compare Oelker & Tutz, 2017)

$$J_{jk}(\beta_{jk}) \approx J_{jk}(\beta_{jk}^{(t)}) + \frac{1}{2} \left(\beta_{jk}^{\top} \mathbf{P}_{jk}(\beta_{jk}) \beta_{jk} + (\beta_{jk}^{(t)})^{\top} \mathbf{P}_{jk}(\beta_{jk}^{(t)}) \beta_{jk}^{(t)} \right),$$

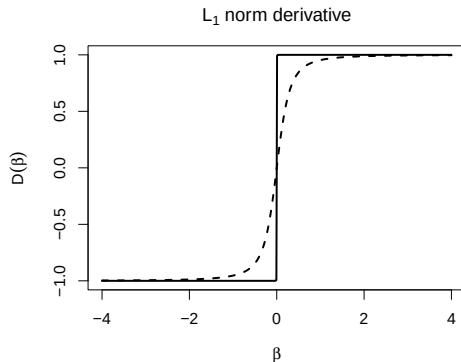
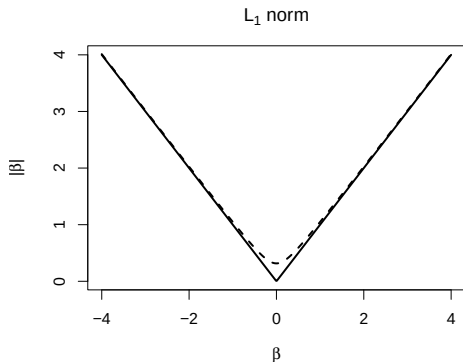
with

$$\mathbf{P}_{jk}(\beta_{jk}^{(t)}) = q'_{jk} \left(\left\| \mathbf{a}_{jk}^{\top} \beta_{jk}^{(t)} \right\|_{N_{jk}} \right) \cdot \frac{D_{jk}(\mathbf{a}_{jk}^{\top} \beta_{jk}^{(t)})}{\mathbf{a}_{jk}^{\top} \beta_{jk}^{(t)}} \cdot \mathbf{a}_{jk} \mathbf{a}_{jk}^{\top}.$$

E.g., $\|\beta\|_1 = |\beta|$ is approximated by $\sqrt{\beta^2 + c}$, hence, IWLS based updating functions $U_{jk}(\cdot)$ are relatively easy to implement.

L1-Type Penalization

Example of the approximation of the L_1 norm.



Usually setting the constant to $c \approx 10^{-5}$ works well.

Application

Install and load.

```
R> install.packages("gamlss2",  
+   repos = c("https://gamlss-dev.R-universe.dev", "https://cloud.R-project.org"))  
R> library("gamlss2")
```

Set up a model formula.

```
R> f <- ~ la(~L+R+P1+L1+R1+P2+E2+C2+G2+O2+P3+P7+L7, type=3) +  
+   s(month, bs="cc") + s(wday, bs="cc", k=5) + s(hour, k=20, bs="cc")  
R> f <- rep(list(f), 4)  
R> f[[1]] <- update(f[[1]], P ~ .)
```

Estimate model.

```
R> b <- gamlss2(f, data = df$train, family = JSU,  
+   add_ridge = TRUE, criterion = "BIC")
```

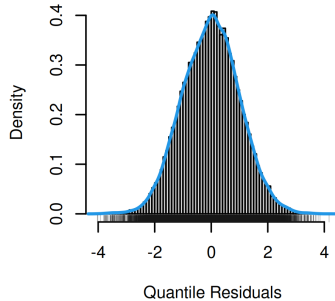
Model summary.

```
R> summary(b)
```

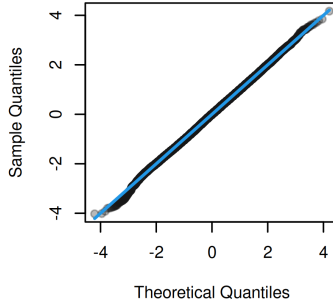
Residuals

```
R> plot(b, which = "resid")
```

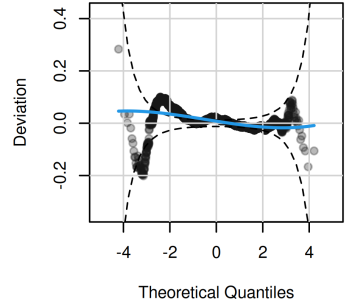
Histogram and Density



Normal Q-Q Plot

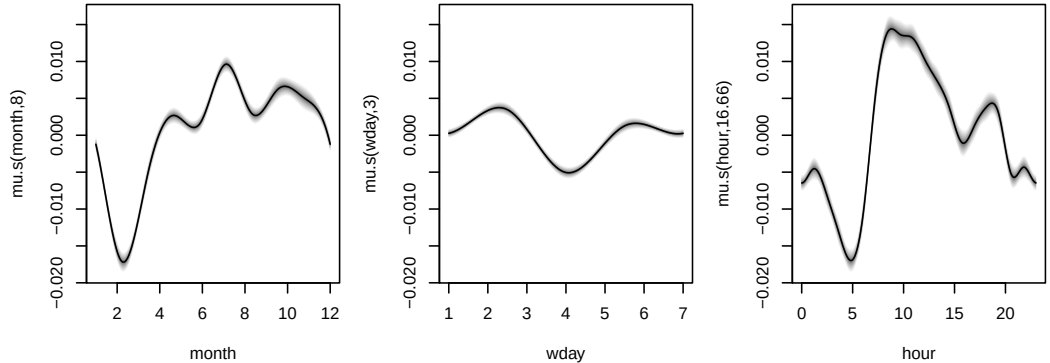


Worm Plot



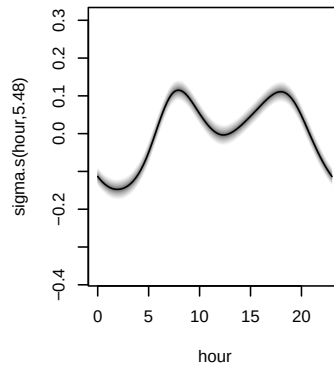
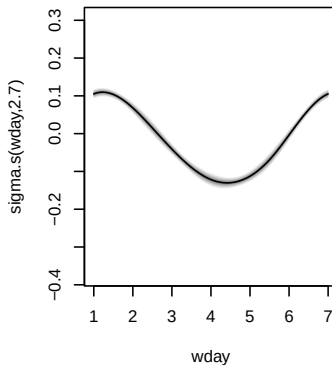
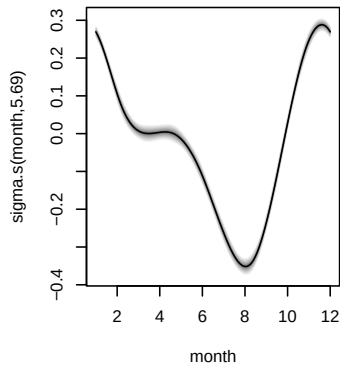
Effects

```
R> plot(b, model = "mu")
```



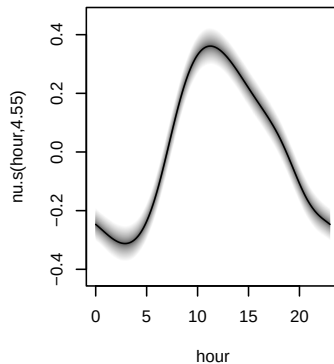
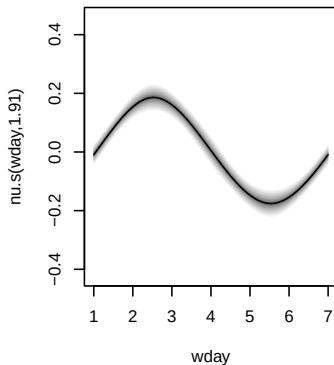
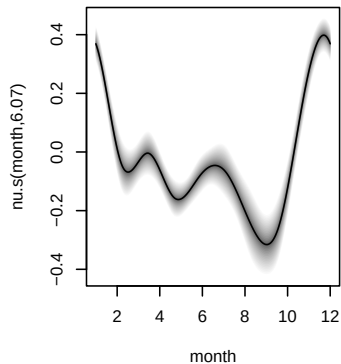
Effects

```
R> plot(b, model = "sigma")
```



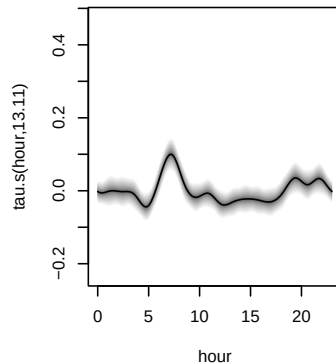
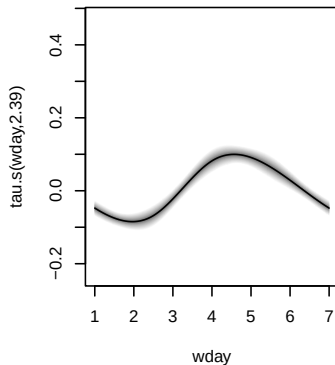
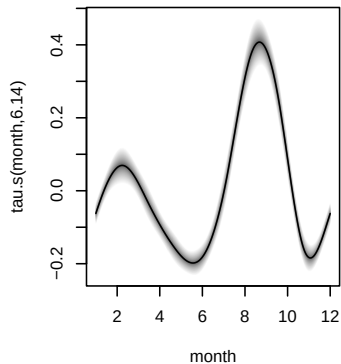
Effects

```
R> plot(b, model = "nu")
```



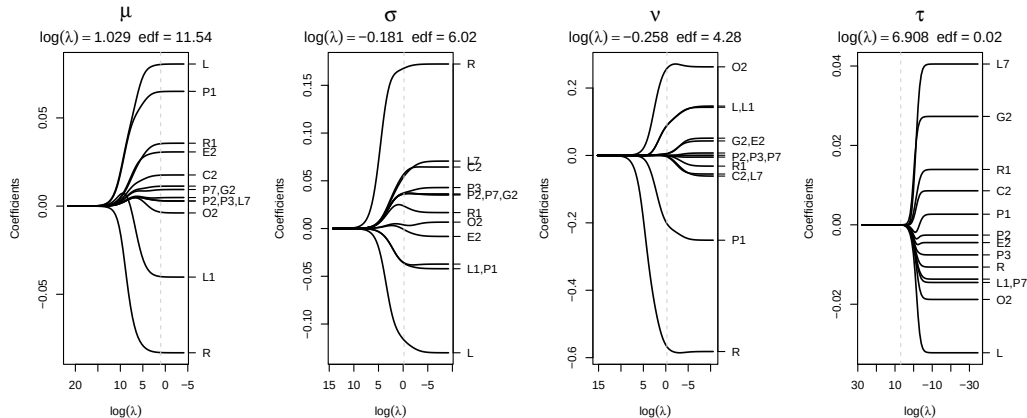
Effects

```
R> plot(b, model = "tau")
```



Coefficients

```
R> plot_lasso(b, which = "coefficients")
```



Predictions

Predict parameters.

```
R> par <- predict(b, newdata = df$test)
```

Predict quantiles.

```
R> qu <- c(0.01, 0.1, 0.5, 0.9, 0.99)
```

```
R> p <- NULL
```

```
R> for(i in qu) {
```

```
+   p <- cbind(p, family(b)$q(i, par) * 100)
```

```
+ }
```

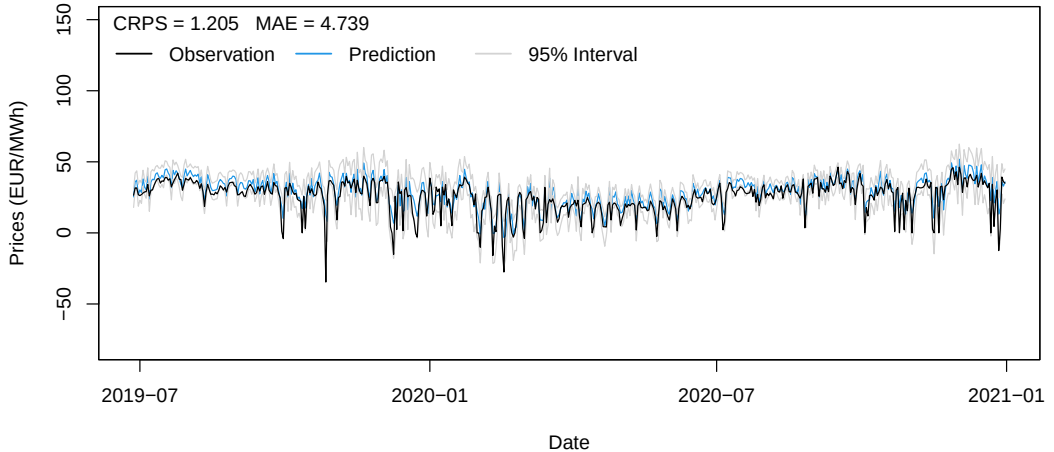
```
R> colnames(p) <- paste0(qu * 100, "%"); rownames(p) <- df$test$time
```

```
R> print(head(p))
```

	1%	10%	50%	90%	99%
2019-06-27 00:00:00	17.18461	23.70882	27.80250	30.93435	34.71296
2019-06-27 01:00:00	15.12706	21.57546	25.62057	28.71319	32.44268
2019-06-27 02:00:00	14.75880	21.21844	25.24996	28.28070	31.88446
2019-06-27 03:00:00	13.73722	20.18522	24.22360	27.29841	30.99556
2019-06-27 04:00:00	13.94446	20.42561	24.43978	27.55404	31.44244
2019-06-27 05:00:00	15.49339	21.95250	25.99306	29.34135	33.79201

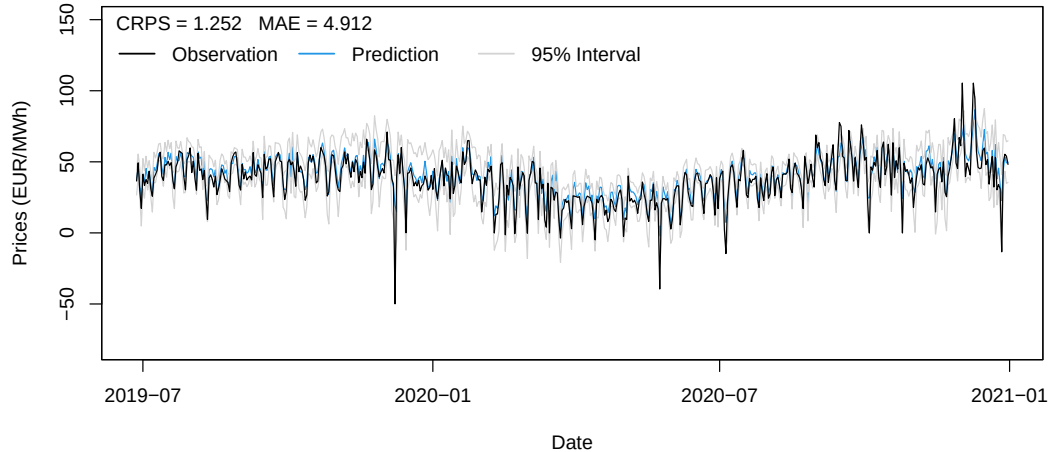
Hourly Prices

Hour 1, forecast:



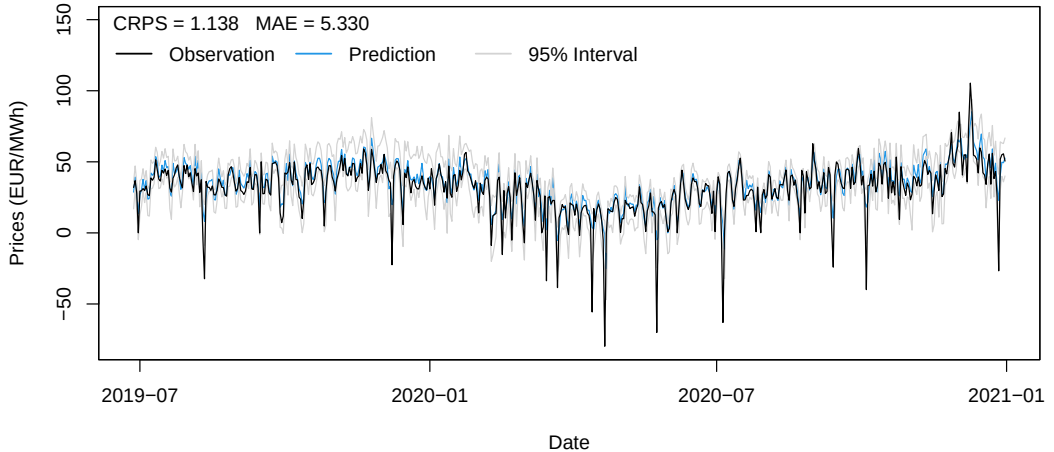
Hourly Prices

Hour 9, forecast:



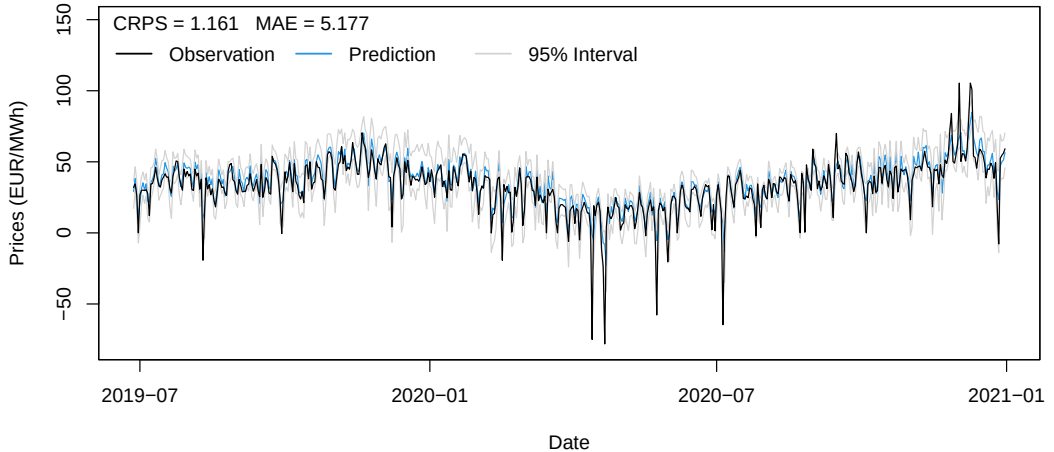
Hourly Prices

Hour 12, forecast:



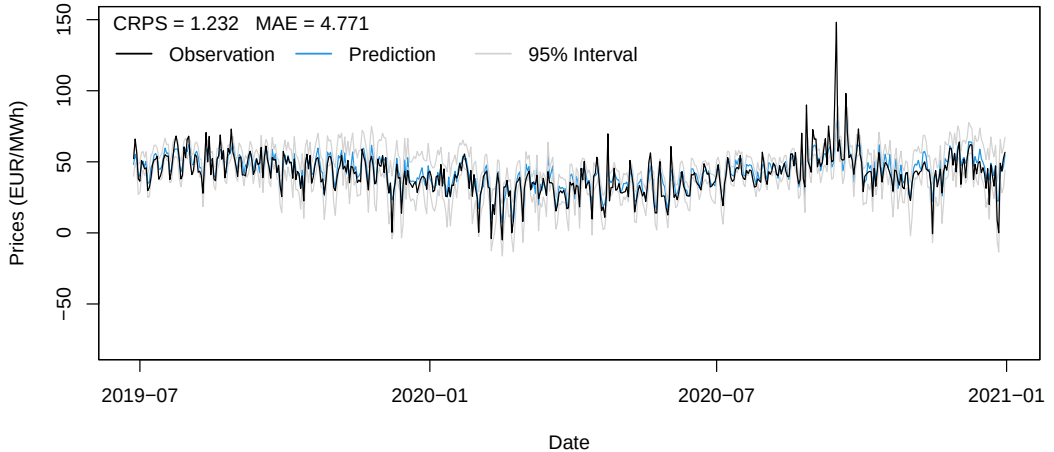
Hourly Prices

Hour 16, forecast:



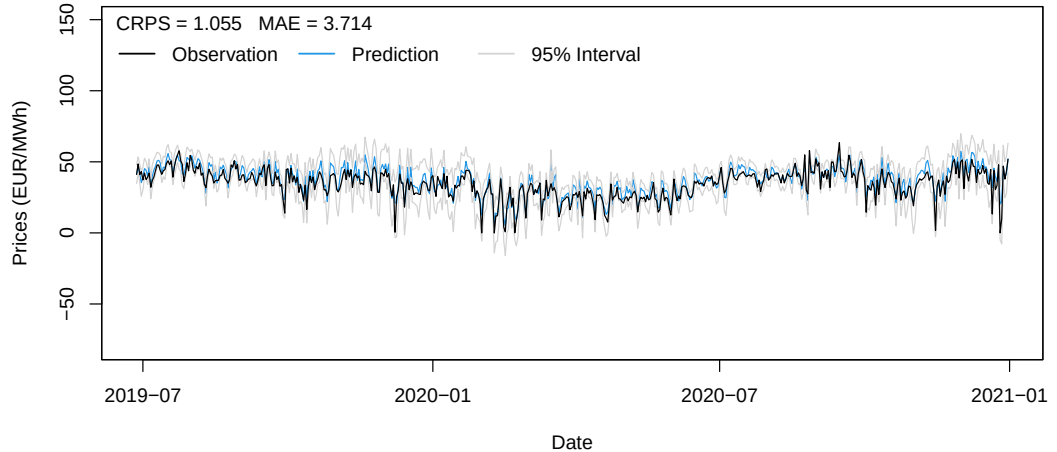
Hourly Prices

Hour 20, forecast:



Hourly Prices

Hour 22, forecast:



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